

03_FOG_SPEED_RESPONSE_RESULTS

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EVIDENCE PROVENANCE: [INJECTED + CLOSED-LOOP]

Time-series logging of weather station fog injection, digital twin state update, and speed governor response.

1. Step-by-Step Telemetry Trace

Time	Visibility	Fog Factor	Actual Speed	Safe Speed	State	PWM
0.0 s	1000 m	1.00	0.80 m/s	0.80 m/s	NORMAL	140
3.0 s	500 m	0.75	0.79 m/s	0.60 m/s	CAUTION	105
6.0 s	250 m	0.50	0.61 m/s	0.40 m/s	SLOW DOWN	70
9.0 s	100 m	0.30	0.41 m/s	0.24 m/s	SLOW DOWN	42
12.0 s	30 m	0.10	0.23 m/s	0.08 m/s	SEVERE STOP	14
15.0 s	1000 m	1.00	0.10 m/s	0.80 m/s	NORMAL	140

2. Dynamic Closed-Loop Metrics

- **Computation Latency:** 4.8 ms average to solve 5-constraint safe speed limit.
- **Clamp Activation:** 14.2 ms from fog change receipt to HMI alert dispatch.
- **Actuator Settling:** 0.65 s smooth deceleration without motor hunting or wheel lock.